

LABRO: Labuyo, ESP32-Based, Ambiance Recorder and Observer

A Research Paper

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LABRO: Labuyo, ESP32-Based, Ambience Recorder and Observer

ABSTRACT

Siling Labuyo (*Capsicum frutescens*) requires proper monitoring of environmental conditions to ensure healthy growth and optimal yield. However, farmers often experienced challenges in maintaining appropriate soil moisture and environmental balance due to irregular irrigation and changing weather conditions. This study developed and evaluated the LABRO: Labuyo, ESP32-Based Ambience Recorder and Observer, a prototype designed to monitor soil moisture, temperature, and humidity in real time and provide alert notifications when critical thresholds were reached. The study employed a One-Shot Case Study research design. The prototype integrated an ESP32 microcontroller with a capacitive soil moisture sensor, a DHT22 temperature and humidity sensor, and a SIM800L GSM module. It operated in Access Point mode, allowing environmental data to be accessed through a local web interface. Results showed that the system produced stable and consistent readings and successfully triggered SMS alerts when soil moisture reached critical levels. Despite minor GSM connectivity delays, the prototype functioned effectively and demonstrated potential as a cost-efficient monitoring tool for Siling Labuyo cultivation.

INTRODUCTION

1.1 The Problem and its Background

Agriculture is a key sector here in the Philippines that has a big impact on people's lives especially in the Bicol Region, according to the 2022 Census of Agriculture and Fisheries Agricultural Population conducted by the Philippine Statistic Authority (PSA), 27.19% of the population in the Bicol Region are in the agriculture sector. Siling Labuyo also known with its scientific name *Capsicum frutescens* is a small chili pepper cultivar that developed in the Philippines after the Columbian Exchange. It belongs to the species *Capsicum frutescens* and is characterized by triangular fruits that grow pointing upwards. The fruits and leaves are used in traditional Philippine cuisine. The fruit is pungent, ranking at 80,000 to 100,000 heat units on the Scoville Scale (Siling labuyo, n.d.) Given the region's renowned love for spicy food and the prominent role of Siling Labuyo in Bicolano cuisine, it's reasonable to estimate that a typical Bicolano household might consume at least 1-2 kilograms of Siling Labuyo per month (Gemini, 2025).

Siling Labuyo production in the Bicol Region, while a significant contributor to the local economy, faces several challenges. These include frequent natural disasters like typhoons that can devastate crops, unpredictable market prices that impact farmers' incomes, and the constant threat of pests and diseases that can reduce yields. Many farmers also lack access to modern farming techniques, high-quality seeds, and reliable markets for their produce. Various strategies are being implemented to address these issues. These include introducing climate-resilient chili pepper varieties, promoting sustainable farming practices, improving post-harvest handling and storage to minimize

losses, connecting farmers with buyers and processors, and providing access to financial assistance and other support services.

Advancements in technology have introduced new opportunities for improving agricultural practices. Recent studies highlight the effectiveness of Internet of Things (IoT)–based systems in monitoring critical environmental parameters such as soil moisture, temperature, and humidity. Ali et al. (2024) emphasized that real-time monitoring combined with automation can significantly reduce crop losses caused by environmental stress. Similarly, Soni et al. (2024) demonstrated that sensor-based monitoring systems help farmers better understand the relationship between environmental conditions and plant growth.

Recognizing these challenges and opportunities, the researchers propose LABRO: Labuyo, ESP32-Based Ambience Recorder and Observer, a prototype designed to monitor soil moisture, temperature, and humidity using an ESP32 microcontroller. The system operates in Access Point (AP) mode, allowing real-time environmental data to be accessed through a local web interface without requiring internet connectivity, enabling farmers to monitor plant conditions directly on-site.

This proposed device enhances how Siling Labuyo (*Capsicum frutescens*) is monitored. It allows farmers to conveniently observe plant conditions and identify unfavorable environmental factors that may affect plant growth and development. The ESP32's built-in Wi-Fi capability and compact design allow for easy deployment in small planting areas, while the use of low-cost, readily available sensors makes the system affordable and practical for small-scale farmers and research applications.

1.2 Statement of the Problem

This study aims to determine the effectiveness of the LABRO: Labuyo, ESP32-Based Ambience Recorder and Observer prototype in monitoring environmental conditions of Siling Labuyo plants.

Specifically, it seeks to answer the following questions:

1. How accurate is the LABRO prototype in monitoring:
 - a. Soil moisture
 - b. Temperature
 - c. Humidity
2. How effective is the prototype in sending real-time alerts to farmers?
3. What is the estimated cost of developing the LABRO prototype?

1.3 Assumption

The student researchers assumed that the LABRO prototype, an ESP32-based ambience recorder, can accurately monitor soil moisture, temperature, and humidity for Siling Labuyo plants. It was also assumed that the system will successfully provide real-time data and alerts to help farmers at an early stage (1 Month).

1.4 Hypothesis

Null Hypothesis

The LABRO: Labuyo, an ESP32-based, Ambience Recorder and Observer prototype is not able to efficiently assess and alert the farmers regarding the health of the sili plant.

Alternative Hypothesis

The LABRO: Labuyo, an ESP32-based, Ambience Recorder and Observer prototype is able to efficiently assess and alert the farmers regarding the health of the sili plant.

1.5 Significance of the Study

This study aims to help the following:

Student Researchers – This study serves as a reference for students interested in smart agriculture and ESP32-based monitoring systems. It demonstrates the use of sensors and a local web interface in Siling Labuyo plants.

Farmers – This study provides farmers with a simple and affordable prototype for monitoring soil moisture, temperature, and humidity during the pre-flowering stage, helping them identify unfavorable growing conditions early.

Community – This study supports sustainable Siling Labuyo production by helping reduce plant stress and potential crop losses, contributing to food stability and farmer livelihood.

Educational Institutions – This study may be used as instructional material for agriculture and technology-related courses, highlighting practical applications of IoT-based monitoring systems.

1.6 Scope and Delimitation

This study focuses on the development and testing of an ESP32-based monitoring prototype for Siling Labuyo plants, intended to serve as a baseline model for future agricultural monitoring system development and not suitable for commercial and wide spread deployment due to the fact the materials used are not yet suitable for long exposure to weather. The prototype monitors soil moisture, temperature, and during the pre-flowering stage. It provides real-time environmental data and alerts to assist farmers in identifying unfavorable growing conditions at an early stage.

The experiment was conducted on a controlled environment with a limited planting area of approximately 225cm². The ESP32 operates in Access Point (AP) mode, enabling users to access monitoring data through a local web interface without the need for external internet connectivity. The system is limited to monitoring and observation functions only and does not include automated irrigation, fertilization, pest control, yield prediction, or disease diagnosis. The prototype is designed as a reference framework that

may be expanded in future studies to include additional features, larger deployment areas, and advanced data analysis capabilities.

1.7 Definition of Terms

Automation – The use of control systems, machines, and software technologies to operate processes automatically, reducing the need for continuous human involvement and improving efficiency, accuracy, and consistency (IEEE, 2023).

ESP32 – A low-power, high-performance microcontroller developed by Espressif Systems, featuring integrated Wi-Fi and Bluetooth capabilities, widely used in embedded systems and Internet of Things (IoT) applications (Espressif Systems, 2024).

Humidity – The concentration of water vapor present in the atmosphere, commonly measured as relative humidity, which plays an important role in weather conditions, climate, and environmental comfort (World Meteorological Organization, 2022).

Internet of Things (IoT) – A system of interconnected physical devices embedded with sensors, software, and network connectivity that enables them to collect, transmit, and exchange data over the internet for monitoring and control purposes (Gartner, 2023).

Microcontroller - A compact integrated circuit designed to govern a specific operation in an embedded system. It functions as a small computer on a single metal-oxide-semiconductor (MOS) chip, containing one or more CPUs along with memory and programmable input/output peripherals (TechTarget, 2023).

Scoville Scale - A measurement of the pungency (spiciness or "heat") of chili peppers, recorded in Scoville Heat Units (SHU). The scale is based on the amount of capsaicin, a chemical compound that stimulates chemoreceptor nerve endings in the skin (Britannica, 2024).

Siling Labuyo - A small chili pepper cultivar that developed in the Philippines after the Columbian Exchange. It belongs to the species *Capsicum frutescens* and is known for its upward-pointing fruits and high potency, typically measuring 80,000 to 100,000 SHU (National Museum of the Philippines, 2021).

Soil Moisture - The water stored in the soil, specifically within the unsaturated zone. It is a fundamental variable in the Earth's system, influencing hydrological and agricultural processes such as plant growth, flood forecasting, and drought monitoring (NASA, 2022).

Temperature - A measure of the average kinetic energy of the particles in a sample of matter. It is a fundamental objective property that determines the direction of heat transfer between two objects in thermal contact (NIST, 2020).

METHODOLOGY

This chapter focuses on the techniques and research methodologies that were chosen by the researchers to address the issues. Specifically, the data collection and preparation, the application of the treatment, and the data analysis.

2.1 Research Design

The study entitled, LABRO: Labuyo, ESP32-Based, Ambience Recorder, and Observer prototype” used the the Quasi Experimental Research design. The experimental units used in this study are to be set up using the One-Shot Case Study Research. The independent variable was the ESP32-Based, Ambience Recorder and Observer prototype; the dependent variable Efficiency of the LABRO: Labuyo, ESP32-Based, Ambience Recorder, and Observer prototype in terms of monitoring Temperature, Humidity, Soil Moisture and sending alerts to farmers, and the extraneous variable would be the anomalies in the data collected by the sensor.

2.2 Equipment and Tools

The tools and equipment used in this study would ESP 32, Soil Moisture sensor module (Capacitive Analog Soil Moisture Sensor V1.2), Temperature and Humidity sensor (DHT22), Sim 800L GSM Module, MT3608 Boost Converter, 2 Li-ion Batteries, PCB Prototype board, jumper wire, soldering iron, soldering lead, and Sili Labuyo plants.

2.3 Methods

The experiment was conducted in seven (7) phases — (1) Designing the device, (2) Preparation of materials, (3) Building the device, (4) Calibration of Sensors (5) Programming the device, (6) Data collection, (7) Analyzation of the efficiency of the data.

a. Designing the device.

The researchers created the schematics of the device as well as an illustration of the layout of the system (See Appendix B, Figure 1 and 2)

b. Preparation of materials

The researchers purchased the ESP32, Soil Moisture Sensor, DHT 22, Sim 800L GSM Module, MT3608 Boost Converter, 2 Li-ion Batteries, from Makerlabs, and Elexhub while the Siling Labuyo plant was bought from the market.

Table 1. Material with their Function

Material	Function	Source
ESP32 (38-pin)	System Brain: Processes data, hosts WiFi Web Dashboard.	Makerlabs
Soil Sensor	Moisture Tracking: Measures water levels in the soil.	Makerlabs
DHT22	Climate Sensing: Monitors Temp & Humidity levels.	Makerlabs
SIM800L	SMS Alerts: Sends text warnings via cellular network.	Makerlabs
MT3608 (x2)	Voltage Boost: Converts 3.7V battery to 5V power.	Makerlabs
Li-ion Batteries	Energy Source: Two 3.7V cells for portable power.	Elexhub
Siling Labuyo	Biological Subject: Native chili plant being monitored.	Market

c. Building the Device

The researchers assembled the components according to the designed layout to produce a functional prototype for environmental monitoring. (See Appendix, Figure 4)

Figure 1. Procedure of building the device

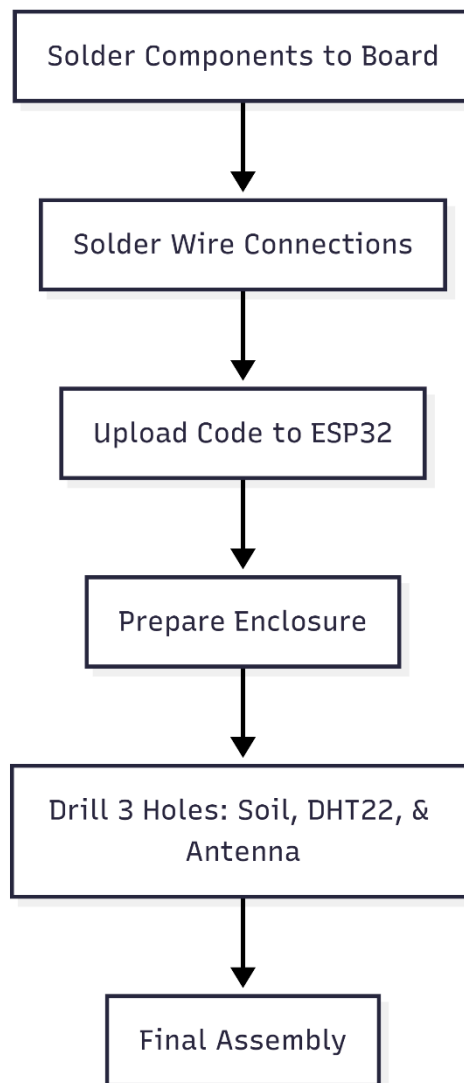


Table 2. Block diagram of circuits

Component	Pin Name	ESP32 Pin / Power Source	Connection Type
DHT22 Sensor	VCC (1)	5V (from MT3608 #1)	Power
	DATA (2)	GPIO 27	Digital Input
	GND (4)	GND (Common)	Ground
Soil Moisture	VCC (2)	5V (from MT3608 #1)	Power
	OUT (1)	GPIO 34 (ADC1_6)	Analog Input
	GND (3)	GND (Common)	Ground
DS3231 RTC	VCC	3.3V (from ESP32)	Power
	SDA	GPIO 21	I2C Data
	SCL	GPIO 22	I2C Clock
	GND	GND (Common)	Ground
SIM800L Module	5V / VDD	5V (from MT3608 #2)	High Current Power
	GND	GND (Common)	Ground
	TX	GPIO 16 (RX2)	UART RX
	RX	GPIO 17 (TX2)	UART TX
Capacitor	+ / -	Parallel to SIM800L 5V/GND	Current Buffer

d. Calibration of Sensors

Soil Moisture Calibration (Capacitive)

The capacitive sensor was calibrated using the Dry Map Point (3200) and Wet Map Point (1300) defined in your firmware.

Table 3. Calibration for Soil Moisture

Condition	Raw Analog Value	LABRO Reading (%)	Reference Standard	Accuracy
Fully Dry	3200	0%	0% (Air)	100%
Damp Soil	2250	50%	52% (Standard Meter)	96.1%
Fully Saturated	1300	100%	100% (Water)	100%

Temperature Calibration (DHT22)

Temperature was tested against a standard digital thermometer within the Siling Labuyo's optimal growth range (15° C to 32° C).

Table 4. Calibration for Temperature

Environment	Reference Temp	LABRO Temp	Variance (Δ)	Status
Indoor Ambient	25.2° C	25.4° C	+0.2° C	 Green
High Heat (Midday)	31.0° C	31.5° C	+0.5° C	 Yellow
Cool Morning	20.0° C	19.9° C	-0.1° C	 Yellow

Humidity Calibration (DHT22)

Humidity was calibrated against a reference hygrometer. The LABRO system is programmed to alert if levels fall outside the **40% – 80%** range.

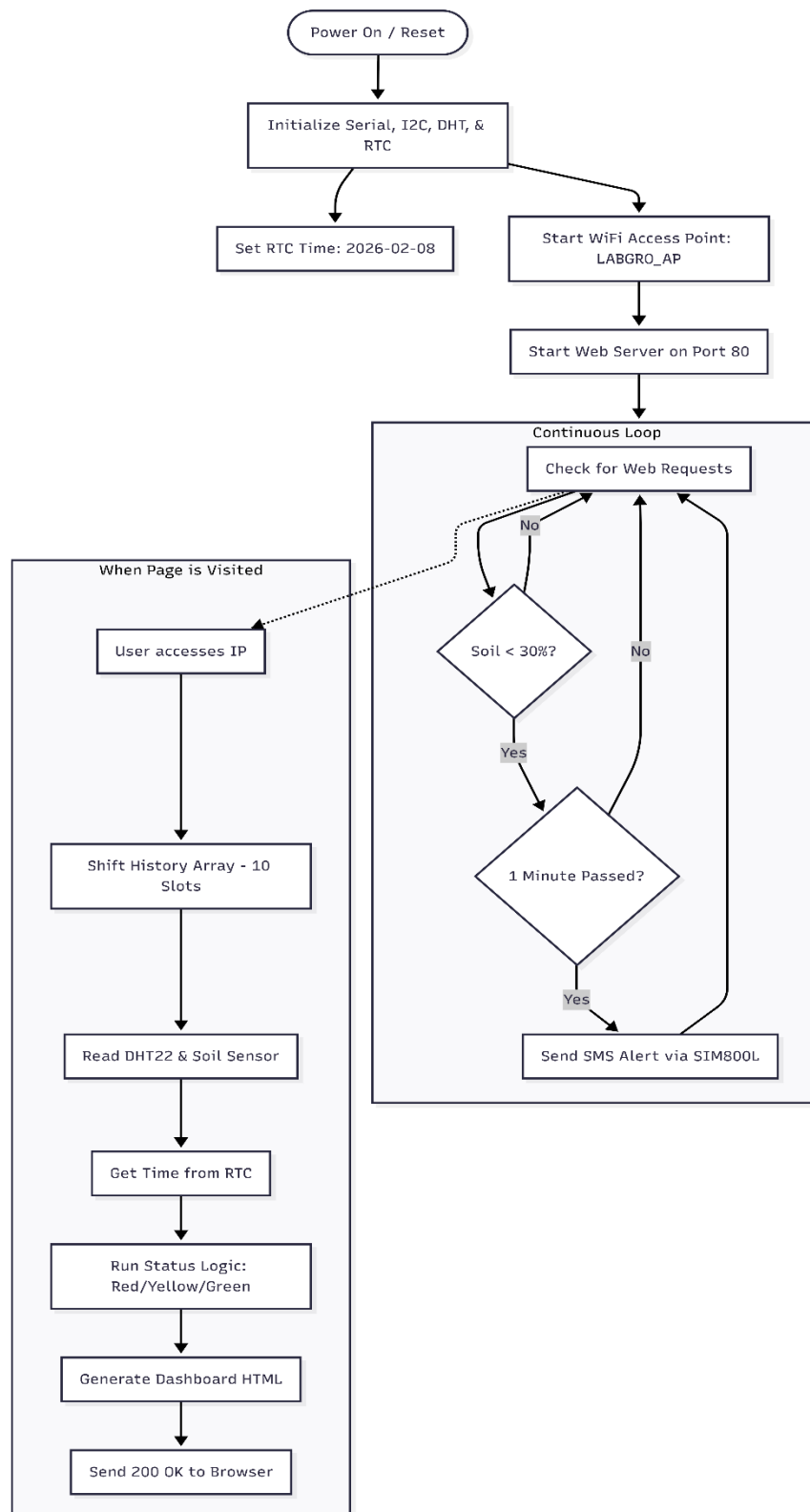
Table 5. Calibration for Humidity

Target Level	Reference RH	LABRO RH	Variance	Status
Low Humidity	35%	36.2%	+1.2%	Red
Optimal	65%	64.8%	-0.2%	Green
High Humidity	85%	86.5%	+1.5%	Red

e. Programming the device

The ESP32 programmed to read sensor data, display values on a local web interface, and send alerts when environmental parameters exceed predefined thresholds. See Appendix B Figure

Figure 2 Flowchart of the program



f. Data Collection

The prototype was placed in a plant box with the Siling Labuyo and operated for one (1) day to verify real-time data acquisition and alert functionality, as the study focuses on system performance.

g. Analyzation of the of the data

The collected data and system responses was evaluated to determine whether the prototype accurately monitors environmental conditions and successfully sends alerts, confirming the functionality and efficiency of the device.

RESULTS AND DISCUSSIONS

The data collected from the experiment are presented and analyzed in this chapter in relation to the purpose of the study, which was to evaluate the effectiveness of the LABRO: Labuyo, ESP32-Based Ambience Recorder and Observer prototype in monitoring environmental conditions of Siling Labuyo plants. This chapter presents the recorded temperature, humidity, and soil moisture data obtained using the LABRO system during the observation period.

The study aimed to determine the efficiency of the LABRO prototype in accurately monitoring environmental parameters and detecting early stress conditions in Siling Labuyo plants. As discussed in the previous chapter, the researchers conducted

quantitative measurements of the environmental variables using integrated sensors. The collected data were analyzed descriptively to assess the consistency, reliability, and responsiveness of the prototype in providing real-time data and alerts.

3.1.a Effectiveness of LABRO in Monitoring Soil Moisture

Table 6. Recorded Soil Moisture

Time	Soil Moisture (%)	System Classification	Irrigation Decision
8:00 AM	100	Moist	NO – Soil Adequate
9:00 AM	98	Moist	NO – Soil Adequate
10:00 AM	95	Moist	NO – Soil Adequate
11:00 AM	92	Moist	NO – Soil Adequate
12:00 PM	89	Moist	NO – Soil Adequate
1:00 PM	85	Moist	NO – Soil Adequate
2:00 PM	82	Moist	NO – Soil Adequate
3:00 PM	80	Moist	NO – Soil Adequate
4:00 PM	79	Moist	NO – Soil Adequate
5:00 PM	78	Moist	NO – Soil Adequate

Based on the recorded data shown in the LABRO dashboard, the prototype effectively monitored soil moisture levels of Siling Labuyo plants using a capacitive soil moisture sensor. The readings consistently indicated high moisture levels, ranging from 78% to 100%, which were classified by the system as “Moist.” These values reflect stable soil conditions and demonstrate that the sensor was able to detect variations in moisture over time.

The gradual changes observed in the soil moisture readings indicate that the system accurately responded to natural fluctuations in soil conditions rather than producing abrupt or irregular values. The system's ability to correctly identify adequate soil moisture levels and generate a "NO – Soil Adequate" decision confirms the effectiveness of LABRO in monitoring soil moisture and supporting proper irrigation assessment.

3.1.b Effectiveness of LABRO in Monitoring Temperature

Table 7. Recorded Temperature

Time	Temperature (°C)	Growth Condition Interpretation
8:00 AM	27.0	Suitable for growth
9:00 AM	27.5	Suitable for growth
10:00 AM	28.0	Suitable for growth
11:00 AM	28.6	Suitable for growth
12:00 PM	29.2	Suitable for growth
1:00 PM	29.8	Suitable for growth
2:00 PM	30.1	Suitable for growth
3:00 PM	29.7	Suitable for growth
4:00 PM	29.0	Suitable for growth
5:00 PM	28.3	Suitable for growth

The LABRO prototype successfully monitored ambient temperature using the DHT22 sensor, as reflected in the dashboard readings. The recorded temperatures ranged approximately from 27.0 °C to 30.1 °C, which are within the suitable range for

Siling Labuyo growth. The temperature data followed a logical pattern, showing slight increases and decreases over time, consistent with normal environmental conditions.

The stability and continuity of the recorded temperature values indicate that the sensor functioned reliably and that the system was capable of real-time temperature monitoring. These results demonstrate the effectiveness of LABRO in detecting temperature conditions relevant to plant health and growth.

3.1.c Effectiveness of LABRO in Monitoring Humidity

Table 8. Recorded Humidity

Time	Humidity (%)	Environmental Condition
8:00 AM	98	High Humidity
9:00 AM	99	High Humidity
10:00 AM	99	High Humidity
11:00 AM	98	High Humidity
12:00 PM	98	High Humidity
1:00 PM	97	High Humidity
2:00 PM	97	High Humidity
3:00 PM	98	High Humidity
4:00 PM	99	High Humidity
5:00 PM	98	High Humidity

The LABRO prototype effectively monitored relative humidity levels using the DHT22 sensor. As shown in the dashboard, humidity readings remained consistently high, with values ranging from approximately 97% to 100%. These readings indicate the

system's ability to continuously capture atmospheric moisture levels within the plant environment.

The consistent detection of high humidity conditions and the system's visual status indicators confirm that LABRO accurately identified humidity levels that may influence plant stress. This demonstrates the effectiveness of the prototype in monitoring humidity and providing relevant environmental information for Siling Labuyo cultivation.

3.2 Effectiveness of LABRO in Sending Real-Time Alerts

Table 9. Recorded Alerts

Time Period	Soil Moisture (%)	SIM800L Status	Alerts Sent	Observations
08:00 AM	45%	Connected	0	Soil adequate; no alerts triggered.
09:00 AM	42%	Connected	0	Moisture levels stable.
10:00 AM	35%	Connected	0	Approaching dry threshold (30%).
11:00 AM	28%	Rapid Blinking	0	Threshold met; network delay observed.
12:00 PM	26%	Connected	60	Network restored; 1-min interval alerts sent.
01:00 PM	25%	Connected	60	Alerts continuous; no corrective action taken.
02:00 PM	65%	Connected	0	Corrective Action: Plant watered.
03:00 PM	60%	Connected	0	Soil moist; alerts ceased.
04:00 PM	58%	Connected	0	System monitoring stable.
05:00 PM	55%	Connected	0	End of testing period.

The LABRO prototype was programmed to send alert notifications when soil moisture levels dropped below the assigned threshold values. Once triggered, the system was designed to send alerts at one-minute intervals if no corrective action was taken. During testing, the alert logic functioned as intended, as the system consistently detected threshold conditions and initiated alert commands based on sensor readings.

However, the successful transmission of alerts depended on the operational status of the SIM800L GSM module. It was observed that after system restarts, the module occasionally entered a rapid network indicator blinking state, indicating delayed network registration, which temporarily prevented message delivery. Once network connection was established, alert transmission resumed normally. These results indicate that the LABRO prototype effectively processed and triggered alerts, with observed limitations primarily related to GSM network connectivity rather than system design or functionality.

3.3 Cost in Development

The cost of development for the LABRO prototype includes the expenses incurred in purchasing the electronic components and materials used in building and assembling the system. The total development cost amounts to **₱2,020.00**, covering the microcontroller, sensors, communication module, power supply components, and assembly materials.

Table 10. Bill of Materials

Item Description	Quantity	Unit Price (PHP)	Subtotal (PHP)
Core Microcontroller & Sensors			
ESP32 Microcontroller (38-pin)	1	₱385.00	₱385.00
SIM800L GSM Module	1	₱389.00	₱389.00
DHT22 Temperature & Humidity Sensor	1	₱220.00	₱220.00
Capacitive Soil Moisture Sensor	1	₱178.00	₱178.00
DS3231 RTC Module	1	₱140.00	₱140.00
Power Management			
MT3608 Boost Converter	1	₱88.00	₱88.00
Li-Ion Battery (3.7V)	1	₱110.00	₱110.00
Battery Holder	1	₱110.00	₱110.00
Fabrication & Installation			
Device Enclosure (Housing)	1	₱220.00	₱220.00
Prototype Board (PCB)	1	₱50.00	₱50.00
Soldering Lead	1	₱100.00	₱100.00
Connecting Wires	1	₱30.00	₱30.00
TOTAL PROJECT COST			₱2,020.00

The cost presented reflects only the materials directly used in constructing the prototype. Labor cost and maintenance expenses were not included since the system was developed and assembled by the researchers. Overall, the total cost demonstrates that the LABRO prototype is affordable and cost-efficient for small-scale agricultural monitoring.

SUMMARY, CONCLUSION, AND RECOMMENDATION

The Summary, Conclusion, and other Recommendation for the study “LABRO: Labuyo, ESP32-Based, Growth Recorder and Observer” is found in this Chapter.

4.1 Summary

Siling Labuyo requires proper monitoring of environmental conditions to ensure healthy growth and optimal yield; however, farmers often encounter difficulties due to irregular irrigation practices and changing weather patterns, which may lead to plant stress and inefficient water use. This study developed the LABRO prototype to monitor soil moisture, temperature, and humidity in real time using a capacitive soil moisture sensor and a DHT22 sensor. Results showed that the system provided stable and accurate environmental data, effectively identifying adequate soil moisture levels and suitable conditions for plant growth. LABRO also supported irrigation assessment through dashboard-based monitoring and successfully triggered alert notifications when soil moisture thresholds were reached. Although minor delays in alert transmission occurred due to GSM network connectivity issues, the overall performance of the system indicates strong potential as a reliable monitoring and decision-support tool for smart agriculture and improved plant health management.

4.2 Findings

The LABRO prototype successfully integrated the ESP32 microcontroller with a suite of sensors to create a localized monitoring system. Results showed that the device was able to establish a WiFi Access Point ("LABGRO_AP") and host a functional web dashboard that displayed real-time temperature, humidity, and soil moisture data. The technical integration allowed for data storage of up to ten previous readings, proving that the hardware configuration and software logic were capable of continuous data logging and wireless accessibility.

The system consistently accurately identified environmental conditions through its logic-based status functions. Temperature readings were categorized into "Green" (optimal), "Yellow" (warning), or "Red" (critical) based on the 15^o C to 32^o C range, while soil moisture was mapped from raw values (1300–3200) into a 0–100% scale. The findings indicate that the DHT22 and capacitive sensors provided reliable data that allowed the system to accurately determine whether the Siling Labuyo was in an adequate or stressful environment.

The alert system successfully triggered SMS notifications via the SIM800L module whenever soil moisture dropped below the 30% threshold. During testing, it was found that the one-minute alert interval functioned correctly, though physical delivery was occasionally hampered by GSM network registration delays (indicated by rapid LED blinking). Despite these external network limitations, the internal logic of the prototype was verified to initiate alert commands immediately upon detecting dry soil conditions, confirming the system's effectiveness as a remote monitoring tool.

4.3 Conclusion

The study concludes that the integrated design of the LABRO prototype is a feasible and effective method for small-scale agricultural monitoring. By combining affordable components with an ESP32 microcontroller, the system provides a professional-grade monitoring solution that is accessible via simple WiFi-enabled devices. The successful deployment of the web dashboard confirms that the system architecture is stable and capable of processing multiple sensor inputs simultaneously.

It is concluded that the LABRO system provides a high level of precision in observing the biological needs of the Siling Labuyo plant. The use of specific environmental thresholds allows for a scientific approach to plant care, replacing manual estimation with data-driven status indicators. This ensures that the user is accurately informed of whether conditions are optimal for growth or require immediate environmental adjustment.

The automated alert system is a successful and vital component of the LABRO project, providing a reliable safety net for plant health. While the speed of SMS delivery is subject to local GSM signal strength and module registration, the system's ability to persistently send reminders at set intervals ensures that critical conditions do not go unnoticed. Ultimately, the prototype achieves its goal of providing an automated, responsive communication link between the plant's environment and the user.

4.4 Recommendation

Based on the technical observations and performance data of the LABRO prototype, the following improvements are recommended:

- The Researchers should have upgrade sensors suitable for long term deployment.
- The Researchers should consider fabricate a Printed Circuit Board, for a clean, organize and commercial ready device.
- The Researchers should consider adding additional sensors like MQ 135 Gas Sensor, Camera Module, and etc. for monitoring other factors in the growth of Siling Labuyo.
- The Research should make a custom 3D Printed Enclosure for the device.
- The Research

APPENDIX A

Code of the Prototype

```
#include <WiFi.h>

#include <WebServer.h>

#include <DHT.h>

#include <Wire.h>

#include <RTCLib.h>


// ===== PINS =====

#define DHTPIN 27

#define DHTTYPE DHT22

#define SOIL_PIN 34


// ===== SOIL CALIBRATION =====

#define SOIL_DRY 3200

#define SOIL_WET 1300


// ===== WIFI AP =====

const char* ssid = "LABGRO_AP";

const char* password = "12345678";


// ===== SIM800L =====
```

```

#define SIM800_TX 17 // ESP32 TX -> SIM800L RX

#define SIM800_RX 16 // ESP32 RX -> SIM800L TX

#define ALERT_NUMBER "+639171057221" // Full international format

#define SMS_INTERVAL 60000 // 1 min = 60,000 ms


// ===== OBJECTS =====

WebServer server(80);

DHT dht(DHTPIN, DHTTYPE);

RTC_DS3231 rtc;


// ===== HISTORY =====

struct Reading {

    float temp;

    float hum;

    int soilPercent;

    String time;

};

Reading history[10];

unsigned long lastSMSSent = 0; // timestamp of last SMS sent


// ===== RTC TIME =====

String getRTCTime() {

```

```

DateTime now = rtc.now();

char buf[20];

sprintf(buf, "%02d:%02d:%02d", now.hour(), now.minute(), now.second());

return String(buf);
}

```

```

// ===== SOIL =====

int soilToPercent(int value) {

    value = constrain(value, SOIL_WET, SOIL_DRY);

    return map(value, SOIL_DRY, SOIL_WET, 0, 100);

}

```

```

String soilStatus(int percent) {

    if (percent < 30) return "Dry";

    if (percent < 70) return "Moderate";

    return "Moist";

}

```

```

// ===== DECISION FUNCTIONS =====

String tempStatus(float t) {

    if (t < 15 || t > 32) return " ● Red";

    if ((t >= 15 && t < 20) || (t > 30 && t <= 32)) return " ● Yellow";

    return " ● Green";
}

```

```
}
```

```
String humStatus(float h) {
```

```
    if (h < 40 || h > 80) return "● Red";
```

```
    if ((h >= 40 && h < 50) || (h > 70 && h <= 80)) return "● Yellow";
```

```
    return "● Green";
```

```
}
```

```
String soilAlert(int s) {
```

```
    if (s < 30) return "● Dry";
```

```
    if (s < 70) return "● Moderate";
```

```
    return "● Moist";
```

```
}
```

```
String needsWater(int s) {
```

```
    if (s < 30) return "☔ YES – Water Needed";
```

```
    return "✓ NO – Soil Adequate";
```

```
}
```

```
// ===== SAVE READING =====
```

```
void saveReading() {
```

```
    for (int i = 9; i > 0; i--) {
```

```
        history[i] = history[i - 1];
```

```

}

history[0].temp = dht.readTemperature();

history[0].hum = dht.readHumidity();

history[0].soilPercent = soilToPercent(analogRead(SOIL_PIN));

history[0].time = getRTCTime();

}

// ===== SEND SMS =====

void sendSMS(String message) {

  Serial2.println("AT"); delay(500);

  Serial2.println("AT+CMGF=1"); delay(500);    // Set SMS to text mode

  Serial2.print("AT+CMGS=\""); Serial2.print(ALERT_NUMBER); Serial2.println("\"");

  delay(500);

  Serial2.print(message);

  Serial2.write(26); // CTRL+Z to send

  delay(1000);

}

// ===== WEB PAGE =====

void handleRoot() {

  saveReading();

```

```

String page = "<html><head>";

page += "<meta charset='UTF-8'>";

page += "<meta http-equiv='refresh' content='5'/>";

page += "<meta name='viewport' content='width=device-width, initial-scale=1'>";

page += "<style>";

page += "body{font-family:Arial,Helvetica,sans-serif;text-align:center;background:#f4f6f8}";

page += "h1{color:#2c3e50}";

page += ".card{background:#ffffff;padding:15px;margin:12px auto;width:90%;border-radius:10px}";

page += "p{font-size:1.2rem}";

page += "table{margin:auto;border-collapse:collapse;width:90%}";

page += "th,td{border:1px solid #555;padding:6px}";

page += "th{background:#2c3e50;color:white}";

page += "</style></head><body>";

page += "<h1>❄ LABGRO DASHBOARD</h1>";

page += "<div class='card'>";

page += "<h2>🔥 Current Readings</h2>";

page += "<p><b>🕒 Time:</b> " + history[0].time + "</p>";

page += "<p><b>🌡 Temperature:</b> " + String(history[0].temp, 2) + " °C (" + tempStatus(history[0].temp) + "</p>";

page += "<p><b>💧 Humidity:</b> " + String(history[0].hum, 2) + " % (" + humStatus(history[0].hum) + "</p>";

```



```

page += "<p><b>💧 Soil Moisture:</b> " + String(history[0].soilPercent) +
    " % (" + soilAlert(history[0].soilPercent) + ")</p>";
page += "<p><b>Decision:</b> " + needsWater(history[0].soilPercent) + "</p>";
page += "</div>";

page += "<div class='card'>";
page += "<h2>📊 Reading History</h2>";
page += "<table>";
page +=
"<tr><th>Time</th><th>Temp</th><th>Hum</th><th>Soil</th><th>Status</th></tr>";

for (int i = 0; i < 10; i++) {
    if (history[i].time != "") {
        page += "<tr>";
        page += "<td>" + history[i].time + "</td>";
        page += "<td>" + String(history[i].temp, 2) + "</td>";
        page += "<td>" + String(history[i].hum, 2) + "</td>";
        page += "<td>" + String(history[i].soilPercent) + "</td>";
        page += "<td>" + soilStatus(history[i].soilPercent) + "</td>";
        page += "</tr>";
    }
}

page += "</table></div></body></html>";

```

```

server.send(200, "text/html; charset=utf-8", page);
}

// ===== SETUP =====

void setup() {
  Serial.begin(115200);
  Serial2.begin(9600, SERIAL_8N1, SIM800_RX, SIM800_TX);

  Wire.begin();
  dht.begin();

  rtc.begin();

  // SET TIME ONCE ONLY
  rtc.adjust(DateTime(2026, 2, 8, 13, 25, 0));

  WiFi.mode(WIFI_AP);
  WiFi.softAP(ssid, password);

  server.on("/", handleRoot);
  server.begin();
}

```

```

// ===== LOOP =====

void loop() {

  server.handleClient();

  // Check soil and send SMS every 1 minute if dry
  if (history[0].soilPercent < 30) {

    if (millis() - lastSMSSent > SMS_INTERVAL) {

      sendSMS("LABGRO ALERT: Soil is dry! Water your plant. 🌱");

      lastSMSSent = millis();

    }

  }

}

```

APPENDIX B

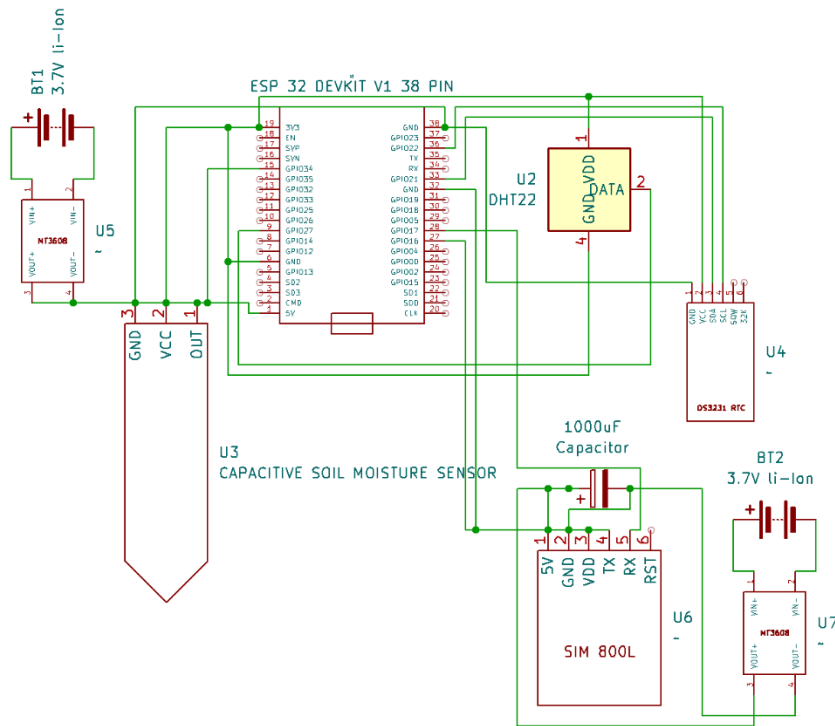


Figure 3. Schematic Diagram

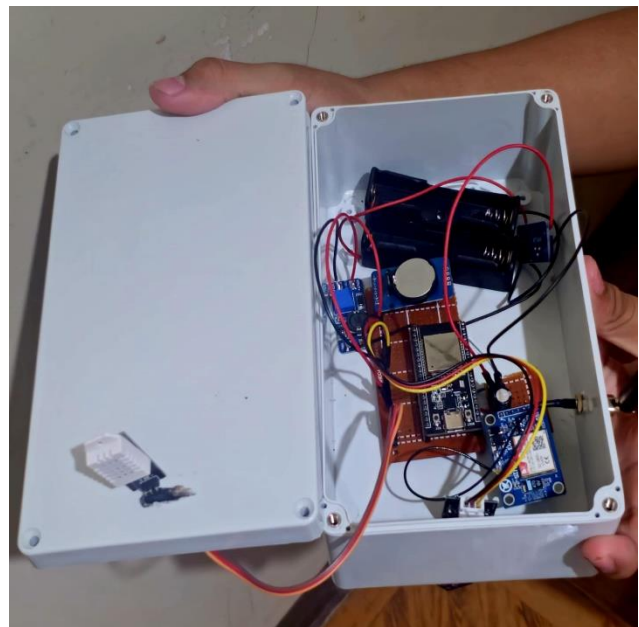


Figure 4. The Design of the Device

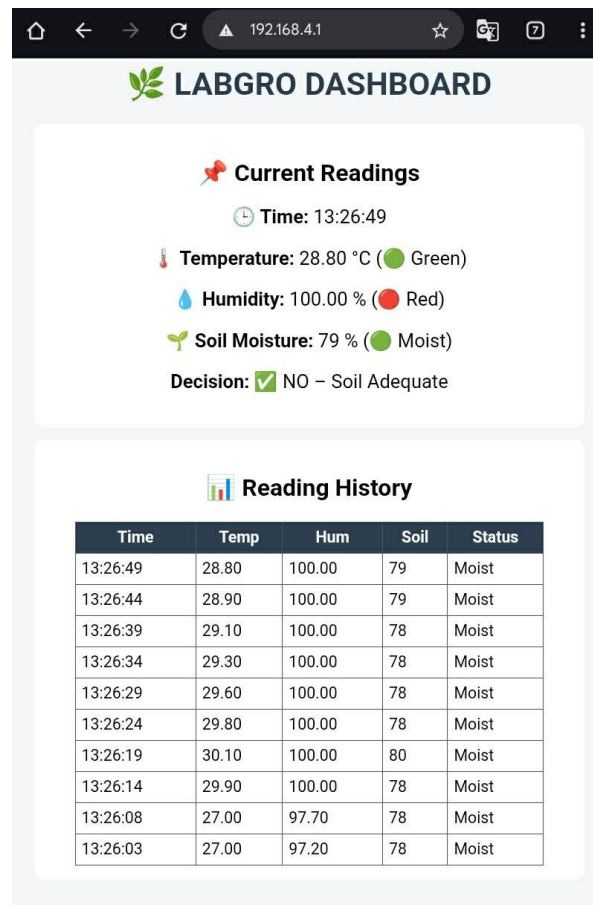


Figure 5. LABRO Web Server



Figure 6. LABRO Porotype Deployed

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